Power Supply Systems Theory

1.List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Devices That Use SMPS (Switched Mode Power Supply)

1. Laptop Charger: Converts AC power from the wall socket into regulated DC power required to charge the laptop battery and operate the laptop.
2. LED Television: Converts the incoming AC electricity into the required DC voltage to power its internal electronic components.
3. Mobile Phone Charger: Converts high-voltage AC electricity into low-voltage DC power to safely charge the phone.

Devices That Use UPS (Uninterruptible Power Supply)

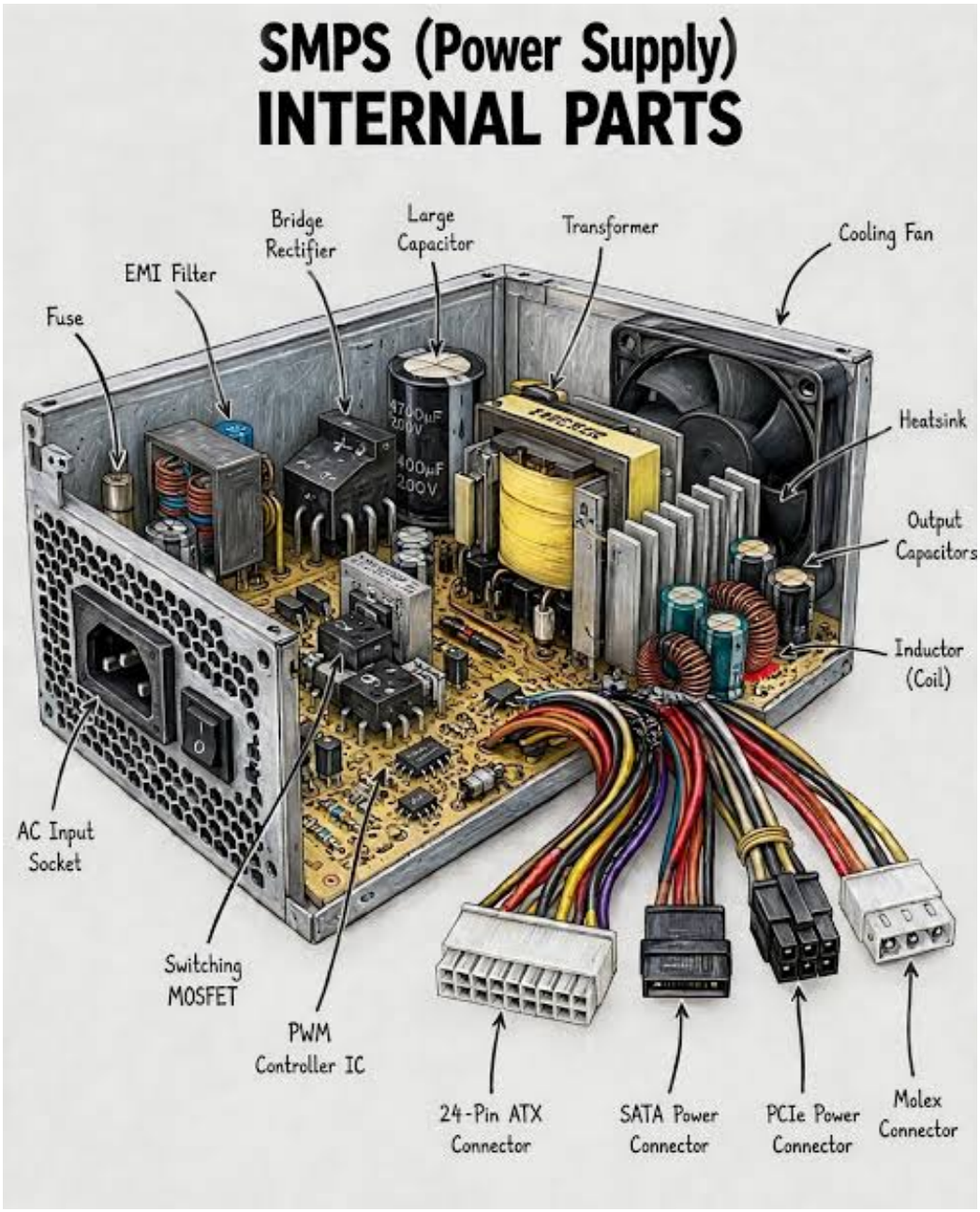
1. Desktop Computer: A UPS provides temporary backup power during a power cut, allowing us to save our work and shut down the computer safely.
2. Wi-Fi Router: A UPS keeps the router running during a power failure so that internet connectivity is not interrupted.

Conclusion: SMPS provides efficient and regulated power to electronic devices, whereas UPS provides backup power during electricity interruptions.

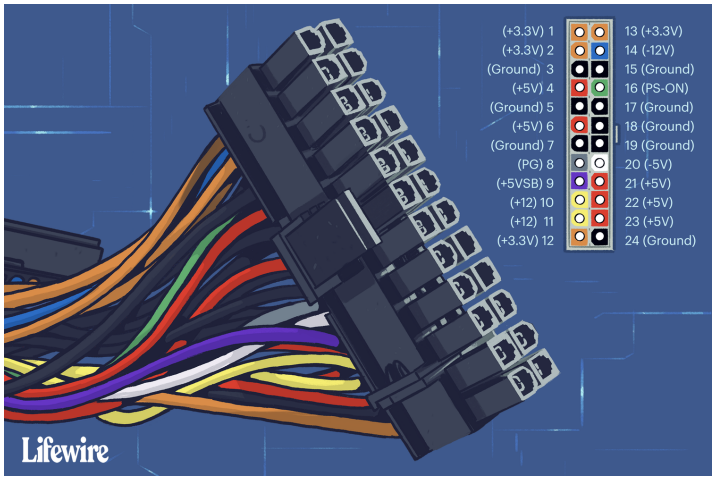
Note: A UPS is generally a separate backup device connected between the electricity supply and the equipment. The devices listed above use UPS backup; they do not necessarily have a UPS built inside them.

**2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.**

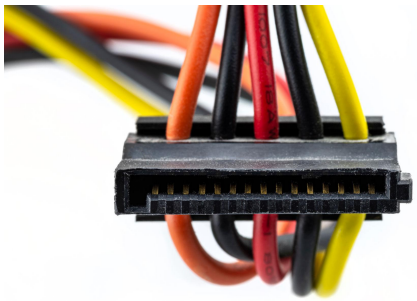


3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



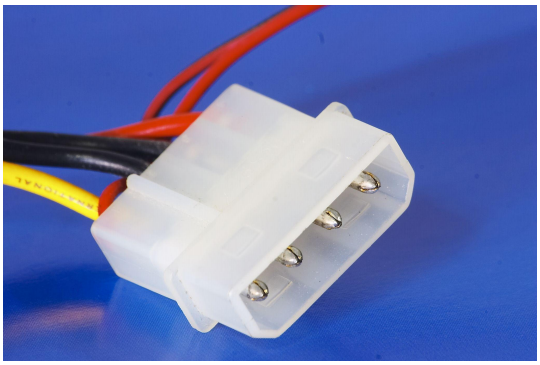
1. ATX 24-Pin Motherboard Connector

- **Name:** ATX 24-Pin Motherboard Connector
  Devices it powers: The main motherboard chipset, onboard controllers, and low-power slots/components.
  Safety Tip: Always ensure the plastic latch/clip clicks securely into place, and make sure the power supply is switched off and unplugged from the wall before plugging or unplugging it to prevent short circuits.



2. SATA Power Connector

- **Name:** SATA Power Connector
  Devices it powers: 2.5-inch and 3.5-inch SATA hard disk drives (HDDs), solid-state drives (SSDs), optical disc drives, and internal accessories like fan controllers or RGB hubs.
  Safety Tip: These flat, L-shaped connectors are directional; never force them in upside down, as doing so can bend the pins or damage the drive's power interface.



3. Molex Connector

- **Name:** Molex 4-Pin Connector
  Devices it powers: Legacy peripherals, older hard drives, optical drives, certain case fans, and RGB/fan hubs.
  Safety Tip: Molex connectors can require a lot of force to pull apart; wiggle them gently side-to-side rather than yanking hard on the thin wires to avoid pulling a wire out of the pin housing.



4. PCIe 6/8-Pin Connector

- **Name:** PCIe 6/8-Pin (or 6+2-pin) Connector
  Devices it powers: Dedicated graphics cards (GPUs).
  Safety Tip: Always press down and hold the release latch on the side of the connector shell before pulling it out of the graphics card to avoid ripping the socket off the PCB.

4.Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Steps to Connect SMPS (Power Supply) in a New Gaming PC

When setting up a new gaming PC, I would follow these steps to connect the SMPS (Switched Mode Power Supply) to the motherboard, graphics card, and storage devices.

Step 1: Prepare the SMPS

- Install the SMPS in the designated cabinet compartment.
- Make sure the computer is switched off and the power cable is disconnected.
- Identify the required power cables: 24-pin ATX, 8-pin CPU, PCIe 6/8-pin, and SATA power cables.

Step 2: Connect the SMPS to the Motherboard

- Identify the PCIe 6-pin, 8-pin (6+2), or 12V-2x6 cable, depending on the graphics card.
- Connect the appropriate cable to the graphics card's power socket.
- Ensure the connector is fully inserted and the locking clip is secured.

Step 4: Connect the SMPS to Storage Devices

- Take the SATA power cable from the SMPS.
- Connect it to the SATA power port on the HDD or SATA SSD.
- If using an NVMe SSD, it does not require a separate SMPS power cable because it gets power directly from the motherboard.

Step 5: Check All Connections

- Check that all power connectors are properly connected.
- Ensure no cables are touching the CPU or cabinet fans.
- Organize the cables properly for good airflow.
- Close the cabinet and connect the power cable.

Step 6: Start the Computer

- Switch on the SMPS and press the computer's power button.
- Check whether the fans, motherboard, and graphics card are receiving power.
- Verify that the computer starts successfully.